Yongan (Luke) Zhang

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Research Interest

- LLM agents for hardware design automation
- Efficient ML systems and software-hardware co-design

Education

- **Georgia Tech** Atlanta, GA, USA
  PhD, Computer Science, Advisor: Prof. Yingyan Lin *Aug. 2023 – May. 2025*
- **Rice University** Houston, TX, USA
  MS, Electrical and Computer Engineering, Advisor: Prof. Yingyan Lin *Jan. 2020 – May 2023*
- **Rice University** Houston, TX, USA
  BS, Electrical and Computer Engineering *Aug. 2015 – May 2019*

Experiences

- **Research Engineer, Meta** Manager: Andrii Gushchyk
  Efficient ML Virtual Telepresence *May 2025 - Present*
  - Develop compiler- and system-level optimizations for efficient on-device ML inference, including operator fusion, kernel scheduling, quantization, pruning, and adaptive runtime techniques across heterogeneous mobile GPU systems for virtual telepresence applications.
- **Research Intern, Meta** Mentor: Dr. Yuecheng Li
  Adaptive Once-for-all model compression *May 2024 - Jan 2025*
  - Designed a once-for-all model compression framework for audio-driven real-time human avatar generation, enabling efficient design space exploration and flexible deployment across GPU latency–memory tradeoff points.
  - Investigated runtime-adaptive model compression as a domain-specific acceleration opportunity, including micro-architectural and compiler support for dynamic tensor shapes and sparsity dispatch.
- **Research Intern, Meta** Mentor: Dr. Yuecheng Li
  Reconfigurable hardware acceleration for VR mobile telepresence pipeline *May 2022 - Dec 2022*
  - Designed a runtime-reconfigurable accelerator architecture for improved compute and memory resource utilization.
  - Developed fine-grained operator scheduling and tiling for efficient model-to-hardware mapping.
  - Built RTL-verified performance modeling for flexible design space exploration across workloads and hardware configs.
  - Implemented an end-to-end compilation flow that auto-generates micro-architecture design and operator scheduling from PyTorch graphs.
- **Ph.D. Intern, PNNL** Mentor: Dr. Ang Li
  Multi-FPGA acceleration for scalable Graph Neural Networks implementation *Jan 2022 – May 2022*
  - Designed a distributed multi-FPGA architecture for scalable GNN inference.
  - Took the design from architecture specification to board-level deployment on production FPGA hardware.

Publications

LLM Agents for Hardware Design Automation

1. **Y. Zhang**, Z. Yu, Y. Fu, C. Wan, Y. Lin, "MG-Verilog: Multi-grained Dataset Towards Enhanced LLM-assisted Verilog Generation", *1st IEEE International Workshop on LLM-Aided Design (LAD), 2024, Best Paper.*
2. **Y. Zhang**, Y. Fu, Z. Yu, K. Zhao, C. Wan, C. Li, Y. Lin, "INVITED: Data4AIGChip: An Automated Data Generation and Validation Flow for LLM-assisted Hardware Design", *The 58th Design Automation Conference (DAC), 2024.*
3. Y. Fu*, **Y. Zhang***, Z. Yu*, S. Li, Z. Ye, C. Li, C. Wan, Y. Lin, "GPT4AIGChip: Towards Next-Generation AI Accelerator Design Automation via Large Language Models", *ACM/IEEE International Conference On Computer Aided Design (ICCAD), 2023.*

Automated Design Frameworks for ML Accelerators

1. **Y. Zhang**, X. Zhang, P. Xu, Y. Zhao, C. Hao, D. Chen, Y. Lin, "AutoAI2C: An Automated Hardware Generator for DNN Acceleration On Both FPGA and ASIC", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.*
2. Y. Zhao, C. Li, Y. Wang, P. Xu, **Y. Zhang**, Y. Lin, "DNN-Chip Predictor: A Multi-grained Graph-based Performance Simulator for DNN Accelerators", *International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2020.*
3. P. Xu, Y. Zhao, C. Hao, X. Zhang, Z. Guan, **Y. Zhang**, Y. Wang, D. Chen, Y. Lin, "AutoDNNchip: An Automated DNN Chip Predictor and Builder for Both FPGAs and ASICs", *ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA), 2020.*

Algorithm–Accelerator Co-Search and Co-Design

1. **Y. Zhang**, H. You, Y. Fu, T. Geng, A. Li, Y. Lin, "G-CoS: GNN-Accelerator Co-Search Towards Both Better Accuracy and Efficiency", *IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2021.*
2. **Y. Zhang**, Y. Fu, W. Jiang, C. Li, H. You, M. Li, V. Chandra, Y. Lin, "DIAN: Differentiable Accelerator-Network Co-Search Towards Maximal DNN Efficiency", *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), 2021.*
3. **Y. Zhang**, A. Banta, Y. Fu, M. John, A. Post, M. Razavi, J. Cavallaro, B. Aazhang, Y. Lin, "RT-RCG: Neural Network and Accelerator Search Towards Effective and Real-time ECG Reconstruction from Intracardiac Electrograms", *The ACM Journal on Emerging Technologies in Computing Systems (JETC), 2021.*
4. Y. Fu, **Y. Zhang**, H. You, Y. Lin, "Auto-NBA: Efficient and Effective Search Over The Joint Space of Networks, Bitwidths, and Accelerators", *The International Conference on Machine Learning (ICML), 2021.*
5. Y. Fu, **Y. Zhang**, C. Li, Z. Yu, Y. Lin, "A3C-S: Automated Agent Accelerator Co-Search towards Efficient Deep Reinforcement Learning", *The 58th Design Automation Conference (DAC), 2021.*
6. Y. Fu, Z. Yu, **Y. Zhang**, Y. Jiang, C. Li, Y. Liang, M. Jiang, Z. Wang, Y. Lin, "InstantNet: Automated Generation and Deployment of Instantaneously Switchable-Precision Networks", *The 58th Design Automation Conference (DAC), 2021.*
7. M. Li, Z. Yu, **Y. Zhang**, Y. Fu, Y. Lin, "O-HAS: Optical Hardware Accelerator Search for Boosting Both Acceleration Performance and Development Speed", *IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2021.*
8. C. Li, Z. Yu, Y. Fu, **Y. Zhang**, Y. Zhao, H. You, Q. Yu, Y. Wang, Y. Lin, "HW-NAS-Bench: Hardware-Aware Neural Architecture Search Benchmark", *The International Conference on Learning Representations (ICLR), 2021.*

Domain-Specific Accelerators and Efficient ML Systems

1. **Y. Zhang**, Y. Li, S. Sarwar, H. Sumbul, Y. Fu, H. You, C. Wan, Y. Lin,"Re-CATA: Real-Time and Flexible Accelerator Design Framework for On-device Codec Avatars", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025.*
2. Z. Ye, Y. Fu, J. Zhang, L. Li, **Y. Zhang**, S. Li, C. Wan, C. Li, S. Prathipati, Y. Lin,"Gaussian Blending Unit: An Edge GPU Plug-in for Real-Time Gaussian-Based Rendering in AR/VR", *IEEE International Symposium on High Performance Computer Architecture (HPCA), 2025.*
3. Y. Fu, Z. Yu, J. Li, J. Qian, **Y. Zhang**, X. Yuan, D. Shi, R. Yakunin, Y. Lin,"AmoebaLLM: Constructing Any-Shape Large Language Models for Efficient and Instant Deployment", *The 38th Conference on Neural Information Processing Systems (NeurIPS), 2024.*
4. Y. Zhao, **Y. Zhang**, Y. Fu, X. Ouyang, C. Wan, S. Wu, A. Banta, M. John, A. Post, M. Razavi, J. Cavallaro, B. Aazhang, Y. Lin, "e-G2C: A 0.14-to-8.31 uJ/Inference NN-based Processor with Continuous On-chip Adaptation for Anomaly Detection and ECG Conversion from EGM", *IEEE Symposium on VLSI Technology and Circuits (VLSI), 2022.*
5. H. You, T. Geng, **Y. Zhang**, A. Li, Y. Lin, "GCoD: Graph Convolutional Network Acceleration via Dedicated Algorithm and Accelerator Co-Design", *IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2022.*
6. H. You, Y. Zhao, Z. Yu, C. Wang, Y. Fu, J. Yuan, S. Wu, S. Zhang, **Y. Zhang**, C. Li, V. Boominathan, A. Veeraraghavan, Z. Li, Y. Lin, "EyeCoD: Eye Tracking System Acceleration via FlatCam-Based Algorithm and Accelerator Co-Design", *IEEE/ACM International Symposium on Computer Architecture (ISCA), 2022.*
7. T. Geng, C. Wu, **Y. Zhang**, C. Tang, C. Xie, H. You, M. Herbordt, Y. Lin, A. Li, "I-GCN: A Graph Convolutional Network Accelerator with Runtime Locality Enhancement through Islandization", *IEEE/ACM International Symposium on Microarchitecture (MICRO), 2021.*
8. H. You, X. Chen, **Y. Zhang**, C. Li, S. Li, Z. Liu, Z. Wang, Y. Lin, "ShiftAddNet: A Hardware-Inspired Deep Network", *Conference on Neural Information Processing Systems (NeurIPS), 2020.*

Awards

- IEEE LAD Best Paper 2024
- 1st & 2nd Place University Demonstration at DAC 2022 & 2023
- Distinction in Research and Creative Work 2019